**LAB SHEET 06: BACK-END INFRASTRUCTURE PREPARATION & SHELL SCRIPTING ENVIRONMENTS**

**Aim**

Demonstrate installation and workspace environment path mappings for Python runtimes, Django frameworks, and Visual Studio Code IDE setups.

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**Task 6.1 — Django Virtual Environment Setup**

**Input / Commands**

# Create project folder

mkdir django_project

cd django_project

# Create virtual environment

python -m venv venv

# Activate virtual environment

venv\Scripts\activate

# Install Django

pip install django

# Save requirements pip

freeze > requirements.txt

**Output** django_project/

│

├── venv/

└── requirements.txt

After activation:

(venv) C:\django_project>

Django installation:

Successfully installed django

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**Task 6.2 — VS Code Python Configuration**

**Input / Settings**

Create a .vscode folder and configure Python settings:

```
{
   "python.defaultInterpreterPath": "venv\\Scripts\\python.exe",
   "editor.formatOnSave": true,
   "editor.tabSize": 4,
   "editor.insertSpaces": true
}
```

**Output**

VS Code uses the virtual environment Python interpreter:

Python Interpreter:

venv\Scripts\python.exe

Indentation:

4 Spaces

Format on Save:

Enabled

The Python workspace is configured according to standard PEP8 indentation requirements.

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**Task 6.3 — Automated Deployment Check**

**Input / check_env.py**

```
import sys
 import os
import django
print("Python Version:", sys.version)
print("Python Path:", sys.executable)
print("Django Version:", django.get_version())
if os.environ.get("PATH"):
print("PATH: Available")
 else:
```

```
    print("PATH: Missing")
print("Environment Check Completed")
```

**Output**

```
Python Version: 3.x.x
Python Path: C:\django_project\venv\Scripts\python.exe
Django Version: 5.x.x
PATH: Available
Environment Check Completed
```

**Final Environment Structure**

```
django_project/
|
├── venv/
├── requirements.txt
├── .vscode/
|   └── settings.json
└── check_env.py
```

**Result**

Python, Django virtual environment, VS Code workspace configuration, and automated environment checks were successfully prepared.